Offline Multi-modal RAG System for Privacy-Preserving Question Answering and Image Retrieval

Capstone Project Report submitted in partial fulfillment.
of the requirement for the degree of
B. E. (Information Technology)

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2025-26

CERTIFICATE OF APPROVAL

For

Capstone Project Report

This is to Certify that

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Have successfully carried out Capstone Project entitled

"Offline Multi-modal RAG System for Privacy-Preserving Question Answering and Image Retrieval"

In partial fulfillment of degree course in

Information Technology

As laid down by University of Mumbai during the academic year 2025-26

Under the Guidance of

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1. Introduction

In today's AI-driven world, information retrieval systems are evolving beyond text to incorporate multiple data types such as images, documents, and structured content. This project presents an Offline Multimodal Retrieval-Augmented Generation (RAG) System built using FastAPI, FAISS, and locally hosted LLM models. The system is designed with a privacy-first architecture, ensuring that all document processing, embedding generation, and inference occur entirely on the user's local machine without any data being transmitted to external servers.

The system enables users to query a knowledge base through both text and image inputs, retrieve semantically relevant information, and generate intelligent, context-aware responses in natural language. A key advancement over conventional RAG implementations is the integration of multilingual support via the paraphrase-multilingual-MiniLM-L12-v2 sentence transformer model, which supports over 50 languages and allows users to submit queries in their native language.

The core of the system lies in the combination of Sentence Transformer embeddings for semantic text representation and FAISS for efficient similarity search across large-scale vector databases. The ingestion pipeline processes PDFs into text chunks and extracts embedded images with surrounding contextual text, which are together stored and indexed for fast retrieval. Additionally, the system features an offline voice input module powered by faster-whisper, enabling speech-to-text transcription without any internet dependency after the initial one-time model download.

When a user submits a query, the system identifies the intent (textual, visual, or combined) using multilingual keyword-based heuristics across 12 languages, fetches the top-k most relevant documents, and leverages a local LLM (served via Jan AI) to provide an enriched and coherent response grounded in retrieved context. Answers include clickable source citations linking back to the originating PDF documents.

The backend, developed in FastAPI, ensures high performance and easy scalability through RESTful endpoints with asynchronous background ingestion. The frontend, built with HTML, CSS, and JavaScript (Jinja2 templates), provides a clean and interactive user interface where users can upload files organised in topic-based folders, enter queries, use voice input, and visualize the system's responses seamlessly.

By combining retrieval and generation with multilingual and multimodal understanding, this project demonstrates a practical implementation of advanced AI techniques for privacy-preserving knowledge augmentation, research assistance, and intelligent search systems — all without requiring an internet connection during operation.

2. Aim and Objectives

Aim

The main aim of this project is to develop an Offline Multimodal Retrieval-Augmented Generation (RAG) System that can understand and process both text and image inputs to deliver intelligent, context-aware, and meaningful responses — entirely on the user's local machine with no data leaving the device. The system combines the strengths of information retrieval and generative AI using tools like FastAPI, FAISS, multilingual sentence embeddings, and a locally hosted LLM.

By integrating these technologies, the project seeks to create a privacy-preserving AI assistant capable of understanding multilingual queries, supporting offline voice input, retrieving relevant data and images efficiently, and generating accurate human-like responses with source citations. This approach bridges the gap between traditional text-based search engines and advanced multimodal, multilingual AI systems, making it suitable for use in education, research, healthcare, legal domains, and enterprise-level knowledge management.

Objectives

1. Multimodal Data Ingestion

- Design a pipeline to handle text documents and PDFs, extracting both text chunks and embedded images.
- Use multilingual sentence transformer embeddings (paraphrase-multilingual-MiniLM-L12-v2) to convert text data into 384-dimensional vector representations.
- Extract images from PDFs using PyMuPDF, capturing surrounding contextual text (captions, nearby paragraphs) for semantic image description.
- Organise uploaded PDFs into named category subfolders (e.g., biology, company-docs) for structured document management.
- Support incremental document management — adding or deleting PDFs triggers automatic re-ingestion without manual intervention.

2. Efficient Retrieval using FAISS

- Implement FAISS for high-speed similarity search across large vector databases.
- Retrieve top-k relevant items (k=15, top 5 text chunks for LLM context) efficiently for any given text or image query.
- Store both text and image embeddings in a unified FAISS index for consistent retrieval.

3. Backend Development using FastAPI

- Build a FastAPI-based backend to manage requests for ingestion, retrieval, response generation, file management, and analytics.
- Implement asynchronous background ingestion so PDF upload and processing do not block the user interface.
- Provide secure API endpoints including path-traversal guards for PDF serving.
- Expose additional endpoints for analytics (/analytics), security auditing (/security-audit), PDF preview (/pdf-preview), and health monitoring (/health).

4. Multilingual Query Support

3. Literature Review

The integration of Retrieval-Augmented Generation (RAG) has transformed the way AI systems handle information retrieval and response generation. Traditional language models like GPT and BERT generate fluent text but often lack factual grounding, leading to hallucinated responses. To address this, Lewis et al. (2020) introduced RAG, combining retrieval-based search with generation models to produce contextually accurate and factually grounded responses.

Recent research has extended RAG to multilingual and multimodal systems. The introduction of multilingual sentence transformers — particularly the paraphrase-multilingual-MiniLM-L12-v2 model developed by Reimers and Gurevych (2019) and extended by the Sentence Transformers library — established efficient cross-lingual embedding representations, enabling semantic search across over 50 languages in a unified vector space. This is a significant departure from earlier systems that required separate models per language.

Earlier multimodal work leveraged CLIP (Contrastive Language-Image Pre-training) by OpenAI (2021), which established a shared embedding space for images and text. While powerful, CLIP embeddings are limited to English text-image matching. The current system adopts a text-centric approach to image retrieval: images are embedded via their extracted surrounding textual context using the multilingual MiniLM model, making image retrieval accessible across languages without requiring raw pixel-level embedding.

For vector storage and similarity search, FAISS (Facebook AI Similarity Search) by Meta AI remains a standard tool, offering efficient large-scale vector indexing and fast nearest-neighbor searches on both CPU and GPU. Integrating FAISS with multilingual sentence embeddings enables rapid retrieval of relevant content across diverse, multilingual document collections.

Offline speech recognition has been advanced significantly by OpenAI's Whisper model family, with faster-whisper providing an optimised CTranslate2-based implementation suitable for local deployment. This aligns with the growing demand for privacy-preserving AI applications where audio data must not leave the user's device.

Frameworks like FastAPI have further simplified the deployment of AI models, offering high performance, async processing, and easy API integration. Projects such as LangChain and LlamaIndex provide structured pipelines for retrieval and generation tasks. This project leverages LangChain's PyPDFLoader and RecursiveCharacterTextSplitter for document ingestion while building custom intent analysis and multilingual query routing on top.

This project builds upon these advancements by combining multilingual FAISS-based retrieval, contextual image extraction via PyMuPDF, offline Whisper voice input, and local LLM inference into a unified privacy-preserving RAG system — extending RAG capabilities to multilingual, multimodal, and voice-enabled real-world knowledge management scenarios.

4. Methodology

The methodology outlines the systematic approach followed in designing, developing, and implementing the Offline Multimodal RAG System. This project combines information retrieval, multilingual natural language processing, offline speech recognition, and local language model inference to create a fully privacy-preserving AI assistant capable of understanding text and image inputs across multiple languages.

The methodology consists of several key phases: Data Ingestion, Multilingual Embedding Generation, Vector Indexing using FAISS, Query Processing with Intent Detection, Response Generation using a local LLM, Backend Integration via FastAPI, and Frontend Development with voice input.

1. System Design Overview

The proposed system architecture is divided into three major layers:

1. Data Layer: Responsible for data ingestion, PDF upload organisation, and storage of extracted text chunks and images.
2. Processing Layer: Handles multilingual feature extraction, embedding generation, and FAISS-based similarity search.
3. Application Layer: Manages user interaction, multilingual query handling, offline voice input, and response generation using FastAPI and a local LLM.

Each layer communicates through well-defined interfaces, ensuring modularity, scalability, and ease of maintenance.

2. Data Ingestion and Preprocessing

The first step in building the RAG system is the ingestion of PDF-based data. Users upload PDFs via the Home page, organised into named category subfolders for structured management.

2.1 Data Collection

Users upload PDF documents or entire folders through the frontend interface at the /upload-files endpoint. Files are saved under uploads/<category>/ subfolders. Upload automatically triggers background ingestion via FastAPI's BackgroundTasks, so the user interface remains responsive during processing.

2.2 Text Extraction

For textual data, the system uses LangChain's PyPDFLoader and RecursiveCharacterTextSplitter to load and chunk PDFs into segments of approximately 500 characters with 50-character overlap. Each text chunk retains metadata including the source PDF path, filename, PDF name stem, and page number for source citation generation.

2.3 Image Extraction with Contextual Text

For image data, the system uses PyMuPDF (fitz) to extract all embedded images from each PDF page. For each image, the system captures surrounding contextual text using bounding box expansion with a 50-pixel margin. If no text is found in the immediate surrounding area, a page-level text fallback captures the first 500 characters of page text. This contextual text, combined with a page-level description, forms the image's semantic representation for

5. Proposed System

System Architecture

The proposed system consists of several interconnected components that work together to provide accurate multimodal responses with multilingual support and full offline privacy:

1. Input Processing Module
   - Accepts queries in text, voice, or combined formats in 50+ languages
   - Performs preprocessing including text tokenisation and language detection
   - Converts voice input to text using the offline Whisper model
2. Feature Extraction Module
   - Generates 384-dimensional embeddings for text data using paraphrase-multilingual-MiniLM-L12-v2
   - Converts PDF images into searchable entries via surrounding contextual text extraction and multilingual embedding
   - Combines all embeddings into a unified FAISS vector store
3. Retrieval Module
   - Performs FAISS similarity search (k=15) against the indexed knowledge base
   - Routes results to text context (top 5 chunks) and image grid (up to 6 images) based on intent
4. Generative Response Module
   - Processes retrieved context with the user query using a local LLM via Jan AI
   - Produces coherent, context-grounded, human-like responses
   - Includes clickable source citations for retrieved documents
5. File Management Module
   - Manages PDF upload, organisation by category, listing, and deletion
   - Triggers automatic background re-ingestion on document changes
6. User Interface Module
   - Home page for document management and ingestion status
   - Chat page for query submission, voice input, text+image response display, and citation viewing

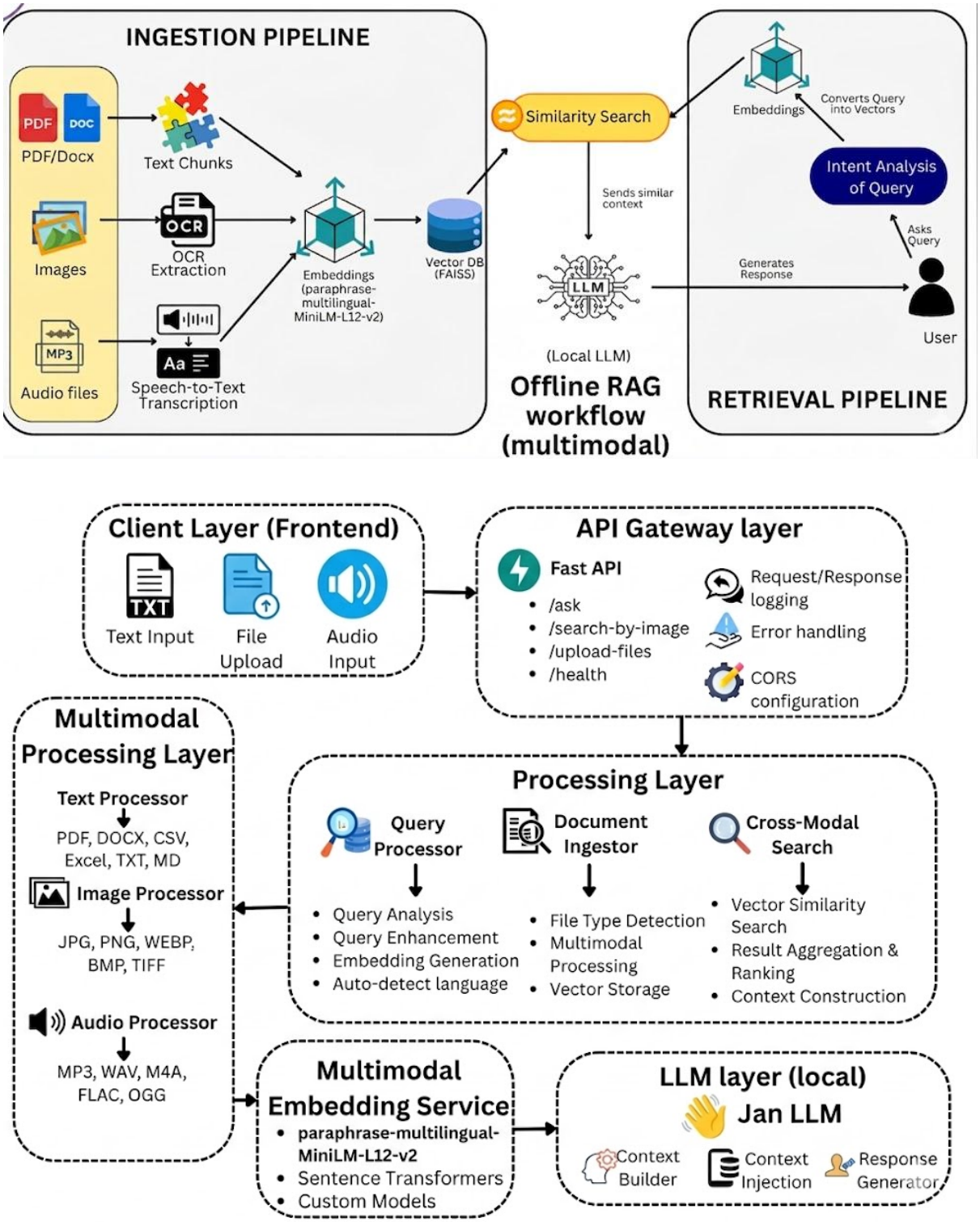

Figure 1: System Architecture

Hardware Requirements

- Processor: Intel Core i5 or equivalent (i7 recommended for faster embedding computation)
- RAM: Minimum 8 GB (16 GB recommended for large model inference and large document sets)

6. Implementation

1. Overview of the System

This system is a fully offline, privacy-preserving document-based AI assistant that can:

- Answer textual questions about uploaded PDF documents in multiple languages
- Retrieve relevant images extracted from PDFs based on semantic similarity
- Provide combined text and image responses for comprehensive answers
- Accept voice queries offline using the locally cached Whisper model
- Display source citations linking answers back to the originating PDF and page

It is built using:

- FastAPI for the backend server with async background task processing
- FAISS for semantic vector search over document embeddings
- Sentence Transformers (paraphrase-multilingual-MiniLM-L12-v2) for multilingual text and image-context embeddings
- Jan AI (local LLM server) for generating natural language answers via OpenAI-compatible API
- faster-whisper for offline speech-to-text transcription
- HTML/CSS/JS frontend (Jinja2 templates) for interactive chat and file management

2. Project Structure

```
offline_RAG_chatbot/ ├── app.py              # FastAPI application (main
entry point) ├── ingest.py           # PDF ingestion & FAISS vector
store builder ├── requirements.txt    # Python dependencies ├──
CLAUDE.md           # Architecture and developer documentation ├──
test.py             # CLI tool for querying vector store ├── bt_print.py
# Serial port utility (unrelated to RAG pipeline) ├── templates/ │   ├──
home.html       # File management UI │   └── index.html      # Chat
interface ├── uploads/            # PDFs organised in category
subfolders ├── vector_store/       # Auto-generated FAISS index +
manifest.json ├── images/             # Auto-extracted PDF images └──
whisper_model/      # Cached Whisper model (first-run download)
```

3. Components

3.1 Backend (app.py)

app.py is the main API service handling the server lifecycle, all API endpoints, query processing, and LLM integration. On startup (lifespan), it loads the FAISS vector store and initialises the local LLM client (openai.OpenAI pointed at http://localhost:1337/v1). Resources are cleaned up on shutdown.

Key query processing steps in the /ask endpoint:

7. Intent analysis: Determines if the query wants text_only, images_only, or both using multilingual keyword dictionaries (12 languages).
8. Language detection: Identifies query language via langdetect with graceful fallback.
9. Query enhancement: Appends contextual terms to improve retrieval for image-intent queries.
10. Vector search: Uses FAISS similarity_search_by_vector (k=15) to find semantically similar entries.

7. Results and Discussion

The implemented Offline RAG Q&A system successfully integrates multilingual text and image retrieval from a local PDF-based knowledge base, demonstrating privacy-preserving semantic search and multimodal response generation. The ingestion pipeline, powered by LangChain and FAISS with multilingual sentence transformer embeddings, effectively processes PDF content by extracting text chunks with page metadata and images with their surrounding contextual text — enabling queries in over 50 languages to retrieve semantically relevant results, including images associated with textual context.

Upon querying, the system dynamically analyses the intent of the input using multilingual keyword-based heuristics across 12 languages to determine whether the user requires text, images, or a combination of both. This multilingual intent detection is a significant advancement over the earlier single-language approach. The FAISS vector store efficiently returns the top-k relevant documents, and the local Jan LLM generates concise, context-aware answers from retrieved text. The frontend seamlessly integrates image results into the chat interface, with interactive features such as image enlargement and clickable citations, ensuring a user-friendly experience.

Testing demonstrates that the system correctly handles diverse query types. Text-only queries (e.g., 'What are the types of fishes?') receive concise textual answers grounded in the uploaded documents. Image-retrieval queries (e.g., 'Show me the image of Bombay-duck fish') return relevant image grids from PDFs. Combined queries (e.g., 'Tell me about dog breeds and show their images') receive both textual explanations and associated image results. Out-of-context queries (e.g., 'Can you tell me the recipe of biryani?') are handled gracefully with a response indicating the information is not available in the uploaded documents — demonstrating effective retrieval grounding and hallucination prevention.

The offline voice input feature, powered by faster-whisper, successfully transcribes queries locally after the one-time model download. The system correctly routes voice-transcribed queries through the same multilingual intent analysis and retrieval pipeline, confirming end-to-end voice query functionality without internet connectivity.

Multilingual query support was verified with queries in multiple languages. The paraphrase-multilingual-MiniLM-L12-v2 model correctly embedded multilingual queries into the same semantic space as the English-embedded document content, enabling cross-lingual retrieval with reasonable accuracy for semantically equivalent queries.

While the system performs well for small to medium-sized knowledge bases, some limitations were observed. Image extraction accuracy depends on PDF structure — images embedded as vector graphics or within complex page layouts may not be extracted accurately. The text-based contextual embedding for images (rather than pixel-level embedding) means that images without surrounding descriptive text may not be retrieved for relevant visual queries. Additionally, the current full-rebuild approach on every ingestion run means that adding a single new PDF to a large document collection requires reprocessing all documents. Future improvements could include incremental index updates, fine-tuning embeddings on domain-specific multilingual data, and integrating an OCR module for scanned PDFs.

Overall, the project demonstrates the feasibility of a fully offline, privacy-preserving multimodal RAG system combining multilingual semantic search, contextual image retrieval, offline voice input, and local LLM inference — providing a robust, interactive knowledge assistant that operates entirely on-device.

8. Conclusion

The developed Offline Multimodal RAG system successfully demonstrates an effective integration of multilingual text and image retrieval from a local PDF-based knowledge repository, operating entirely without internet connectivity during operation. By leveraging multilingual sentence transformer embeddings (paraphrase-multilingual-MiniLM-L12-v2), FAISS-based vector search, contextual image extraction via PyMuPDF, offline voice input via faster-whisper, and local LLM inference through Jan AI, the system is capable of providing precise, context-aware responses to user queries across 50+ languages — while ensuring that all data remains on the user's device.

The project highlights the potential of privacy-preserving multimodal retrieval systems, demonstrating that combining multilingual text embeddings with contextual image extraction can significantly improve the relevance and accessibility of responses across diverse language communities. The addition of offline voice input further lowers barriers for non-keyboard users and establishes a foundation for hands-free document querying. Source citations with clickable links to originating PDFs add transparency and trust to the system's responses.

Despite minor limitations in image caption extraction accuracy for complex PDF layouts and the full-rebuild requirement on every ingestion run, the system performs reliably for small to medium knowledge bases and gracefully handles out-of-context queries without hallucination.

In conclusion, this work validates the feasibility of creating an interactive, privacy-first knowledge assistant that efficiently bridges multilingual semantic text search with contextual image retrieval and offline voice input — providing a robust foundation for future improvements such as incremental index updates, OCR for scanned PDFs, domain-specific embedding fine-tuning, and scalability to larger multilingual document collections.

9. Future Scope

The RAG system developed in this project can be further enhanced in multiple directions:

1. Incremental Index Updates: The current implementation performs a full vector store rebuild on every ingestion run. Implementing incremental updates — adding only new or modified PDFs to the existing index without reprocessing unchanged documents — would significantly reduce ingestion time for large document collections.
2. OCR for Scanned PDFs: Integrating an OCR module (e.g., Tesseract or a neural OCR model) would enable the system to process scanned PDFs and image-only pages that currently produce no extractable text.
3. Domain-Specific Embedding Fine-Tuning: Embedding models and LLMs can be fine-tuned on domain-specific corpora (medicine, law, engineering) to improve retrieval accuracy and response quality for technical subject matter.
4. Support for Additional File Types: Expanding beyond PDFs to include Word documents, Excel sheets, PowerPoint presentations, and web content would make the system more versatile for diverse enterprise use cases.
5. Advanced Image Understanding with Visual Models: Integrating vision-language models (e.g., LLaVA, MiniGPT-4) for pixel-level image understanding would enable richer visual retrieval beyond text-based contextual embedding — particularly for diagrams, charts, and technical figures without adjacent text.
6. Conversational Context Retention: Implementing multi-turn conversation memory would allow the chatbot to reference previous messages within a session, enabling follow-up questions and iterative refinement of answers.
7. Scalability and Cloud-Optional Deployment: While designed for fully local use, adding an optional cloud deployment mode with distributed vector stores and GPU acceleration would support organisations with larger document collections and heavier concurrent usage.
8. Real-Time Index Updates: Implementing file system watchers to automatically detect new, modified, or deleted PDFs and trigger incremental ingestion would eliminate the need for manual re-ingestion triggers.
9. Expanded Multilingual Voice Support: Extending voice input to support language-specific Whisper models (medium, large) for improved transcription accuracy in low-resource languages beyond the base model's capabilities.

10. References

1. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language Models are Few-Shot Learners. Advances in Neural Information Processing Systems, 33, 1877-1901.
2. OpenAI. (2023). OpenAI API Documentation. Retrieved from https://platform.openai.com/docs
3. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP.
4. Johnson, J., Douze, M., & Jegou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 535-547.
5. LangChain Community. (2024). LangChain Community Vectorstores Documentation. Retrieved from https://www.langchain.com
6. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems, 32.
7. Hugging Face. (2024). Sentence Transformers Library. Retrieved from https://www.sbert.net
8. MuPDF. (2024). PyMuPDF Documentation. Retrieved from https://pymupdf.readthedocs.io
9. Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust Speech Recognition via Large-Scale Weak Supervision (Whisper). OpenAI Technical Report.
10. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is All You Need. Advances in Neural Information Processing Systems, 30, 5998-6008.
11. Chen, D., Fisch, A., Weston, J., & Bordes, A. (2017). Reading Wikipedia to Answer Open-Domain Questions. ACL 2017.
12. Wang, S., Pan, Y., & Li, Z. (2022). Enhancing RAG Systems with Contextual Image Understanding. Journal of Artificial Intelligence Research, 74, 105-125.
13. Jan AI. (2024). Jan - Open-source local AI. Retrieved from https://jan.ai
14. faster-whisper. (2024). faster-whisper: Faster Whisper transcription with CTranslate2. Retrieved from https://github.com/SYSTRAN/faster-whisper